

A finite-time convergence of acyclic generically persistent formation in 3-D using relative position measurements

Viet Hoang Pham¹, Minh Hoang Trinh¹ and Hyo-Sung Ahn^{1*}

¹School of Mechanical Engineering, Gwangju Institute of Science and Technology (GIST), Gwangju, Republic of Korea
(vietph@gist.ac.kr, trinhhoangminh@gist.ac.kr, hyosung@gist.ac.kr)

Abstract: This paper studies formation control problem of multi-agent system whose interaction graph is acyclic generically persistent. We design a distributed control law for each agent to maintain its inter-agent distance constraints toward its neighbours. The designed control law uses only local relative position measurements and is proved to guarantee successfully for the convergences of agents to their desired equilibria in finite time, at which the formation of overall group of agents is congruent with a given desired one.

Keywords: Cooperative control, multi-agents system, formation control, persistent graph, acyclic graph, relative position measurement

1. INTRODUCTION

Due to the need for autonomous control of multiple mobile robotic agents sharing the same workspace, cooperative control of multiple autonomous agents is a field that has gained increased attention in the past few years in both the robotics and control communities [1], [2], [3]. With their practical potential in various applications and theoretical challenges arising in formation control, a significant amount of research efforts has been focused on the formation shape control of multi-agent systems. Multi-agent systems are systems consisting of multiple agents or vehicles with several sensors/actuators and the capability to communicate with one another to perform coordinated tasks. Though, no agent can observe/sense the states of all others, each agent is able to measure some geometrically relevant variables involving some of the other agents in the group, to apply control to correct errors in formation shape [4].

Because of its fully decentralized characteristic, distance-based problem is acceptable for the system with large number of agents. Each individual uses only relative positions measurements with respect to its own coordinate system. If the interaction graph of multi-agent system is rigid or persistent, when all inter-agent distance constraints are satisfied, the desired formation of overall group is achieved. When the task to maintain each inter-agent distance constraint belongs to both associated agents, the formation is undirected [5], [6], [7], [11]; otherwise, it is directed [8], [9], [10], [16]. In the research line of distance-based formation control, the desired formation is proved locally asymptotically stable under gradient control law, which is derived from some potential function. However, under these gradient descent control laws, the existence of undesired equilibria associated with a undesired formation shapes can be observed from numerical simulations [5], [16]. In fact, the gradient control law fails to stabilize globally multi-agents formation in general case. Thus, there is an urgent need for design-

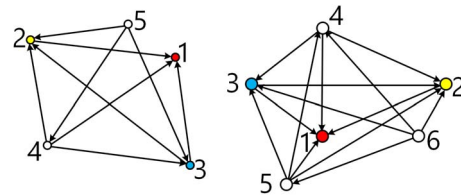


Fig. 1 Example of persistent graph: The left figure is generically persistent in the plane and the right figure is generically persistent in three-dimensional space.

ing a distributed control law to guarantee trajectories of agents converge to their desired equilibria corresponding to the desired formation.

In this paper, we consider formation control problem of a multi-agent system, whose interaction graph is acyclic generically persistent. The control law implemented to each agent uses only the local relative positions of some others with respect to its own coordinate system. The desired formation of overall system is shown to be globally finite-time stable. To the best of our knowledge, this is the first work for globally stabilizing a general multi-agent formation in three-dimensional space. The outline of this paper is given as follows. In Section 2, we give some mathematical backgrounds. The designed control law and stability analysis are provided in Section 3. In Section 4, we show some simulations supporting our analysis and Section 5 provides concluding remarks.

2. PRELIMINARIES

2.1 Some useful lemmas

The finite time stability using a Lyapunov argument was studied in [13], from which we record the following finite-time lemma:

Lemma 1: Consider the system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ where $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{f} : U \rightarrow \mathbb{R}^n$ is a continuous function on an open

neighbourhood $U \subseteq \mathbb{R}^n$ of the origin and $\mathbf{f}(\mathbf{0}) = \mathbf{0}$. Suppose there exist a continuous positive definite function $V(\mathbf{x}) : U \rightarrow \mathbb{R}$, a real numbers $c > 0$ and $\alpha \in (0, 1)$; and an open neighborhood $U_0 \subset U$ of the origin such that $\dot{V}(\mathbf{x}) + c(V(\mathbf{x}))^\alpha \leq 0, \forall \mathbf{x} \in U_0 \setminus \{\mathbf{0}\}$. Then, $V(\mathbf{x})$ approaches 0 in finite time. In addition, the finite settling time function T satisfies $T \leq \frac{(V(\mathbf{x}(0)))^{1-\alpha}}{c(1-\alpha)}$.

Consider function $\text{sig}(x)^\alpha = |x|^\alpha \text{sign}(x)$ where $\alpha > 0, x \in \mathbb{R} \setminus \{0\}$, we have

$$\frac{\partial \text{sig}(x)^\alpha}{\partial x} = \alpha |x|^{\alpha-1}, \quad \frac{\partial |x|^\alpha}{\partial x} = \alpha \text{sig}(x)^{\alpha-1}$$

Lemma 2: [12] Let $x_1, x_2, \dots, x_n \geq 0$, given $\alpha \in (0, 1)$, there holds $(\sum_{i=1}^n x_i)^\alpha \leq \sum_{i=1}^n x_i^\alpha$.

2.2 Generically persistent graph

We use a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ to describe the interaction topology of multi-agent system. $\mathcal{V}$ is the vertex set and each vertex corresponds to an agent; $\mathcal{E}$ is the edge set and each edge $(i, j) \in \mathcal{E}$ corresponds to the inter-agent distance constraint between two agent i, j , which is preserved by agent i . The fact $(i, j) \in \mathcal{E}$ does not imply $(j, i) \in \mathcal{E}$. Let $\mathcal{N}_i = \{j : (i, j) \in \mathcal{E}\}$ be the set of all neighbours of agent i . For simplicity, we label agents such that if $j \in \mathcal{N}_i$, it implies $j < i$. Let $\mathcal{D} = \{\bar{d}_{ij} : (i, j) \in \mathcal{E}\}$ be the set of desired values of inter-agent distance constraints. A position $\mathbf{p}_i$ is fitting for the desired distances if there is no $\mathbf{p}_i^* \in \mathbb{R}^3$ such that

$$\{j \in \mathcal{N}_i : \|\mathbf{p}_i - \mathbf{p}_j\| = \bar{d}_{ij}\} \subset \{j \in \mathcal{N}_i : \|\mathbf{p}_i^* - \mathbf{p}_j\| = \bar{d}_{ij}\}.$$

Then the framework $(\mathcal{G}, \mathbf{p})$, where realization $\mathbf{p} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_N] \in \mathbb{R}^{3N}$, is fitting for the desired distances if all the vertices are at their fitting positions. The persistence of frameworks is defined as follow

Definition 1: Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a directed graph. A framework $(\mathcal{G}, \mathbf{p})$ is persistent if there exists $\epsilon > 0$ such that every realization $\mathbf{p}'$ fitting for the distance set induced by $\mathbf{p}$ and satisfying $\max_{i \in \mathcal{V}} \|\mathbf{p}_i - \mathbf{p}'_i\| < \epsilon$, is congruent to $\mathbf{p}$.

A graph is generically persistent if almost all realization $\mathbf{p}'$ fitting for the distance set induced by $\mathbf{p}$ are congruent to $\mathbf{p}$. A representation is persistent if and only if it is rigid and constraint consistent. The notion "constraint consistent" implies every vertex can reach its fitting position. Details on persistence are found in [14], [15].

2.3 Mathematical background

Let A, B, C, O be the four non-coplanar points in three-dimensional space and assume that every distance between any two points is known, we can calculate the angle $\angle BAC$ between two vectors $\vec{AB}, \vec{AC}$ and the angle $\angle(\Delta_{OAB}, \Delta_{CAB})$ between two planes $\Delta_{OAB}, \Delta_{CAB}$ where Δ_{OAB} is the plane containing A, B, O and Δ_{CAB} is the plane containing A, B, C . Denote $a = \overline{BC}, b = \overline{AC}, c = \overline{AB}, x = \overline{OA}, y =$

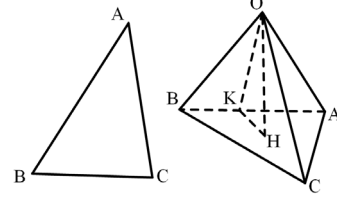


Fig. 2 Mathematical background illustration.

$$\overline{OB}, z = \overline{OC}, \angle BAC = \theta, \angle(\Delta_{OAB}, \Delta_{CAB}) = \angle OKH = \phi.$$

From cosine law, we have $a^2 = b^2 + c^2 - 2bc \cos \theta$. We define the function

$$\theta = d_\theta(b, c, a) = \cos^{-1} \frac{b^2 + c^2 - a^2}{2bc}. \quad (1)$$

Let V be the volume of tetrahedron $OABC$, A_4 is the area of ABC and A_3 is the area of OAB , we have

$$V^2 = \frac{1}{144} [x^2 a^2 (y^2 + z^2 + b^2 + c^2 - x^2 - a^2) + y^2 b^2 (x^2 + z^2 + a^2 + c^2 - y^2 - b^2) + z^2 c^2 (y^2 + x^2 + b^2 + a^2 - z^2 - c^2) - x^2 y^2 c^2 - y^2 z^2 a^2 - x^2 z^2 b^2 - a^2 b^2 c^2], \quad (2)$$

$$A_3^2 = q_3(q_3 - x)(q_3 - y)(q_3 - c), \quad (3)$$

$$A_4^2 = q_4(q_4 - a)(q_4 - b)(q_4 - c). \quad (4)$$

where $q_3 = \frac{1}{2}(x + y + c), q_4 = \frac{1}{2}(a + b + c)$. Denote $l = \overline{OK} = \frac{2A_3}{c}, h = \overline{OH} = \frac{3V}{A_4}, \overline{CH}^2 = z^2 - h^2$, we define the function

$$\phi = d_\phi(c, x, y, a, b, z) = \begin{cases} 180^\circ - \sin^{-1} \frac{h}{l}, & \text{if } \overline{CH} > \frac{2A_4}{c} \\ \sin^{-1} \frac{h}{l}, & \text{if } \overline{CH} \leq \frac{2A_4}{c}. \end{cases} \quad (5)$$

When we know the positions of four points A, B, C, O with respect to any coordinate system, we can calculate angles θ, ϕ from vectors as following functions

$$\theta = f_\theta(\vec{AB}, \vec{AC}) = \cos^{-1} \frac{\vec{AB} \cdot \vec{AC}}{\|\vec{AB}\| \cdot \|\vec{AC}\|}. \quad (6)$$

$$\phi = f_\phi(\vec{OA}, \vec{OB}, \vec{CA}, \vec{CB}) = \cos^{-1} \frac{[\vec{OA} \times \vec{OB}] \cdot [\vec{CA} \times \vec{CB}]}{\|[\vec{OA} \times \vec{OB}]\| \cdot \|[\vec{CA} \times \vec{CB}]\|}. \quad (7)$$

3. DESIGN CONTROL STRATEGY

3.1 Formation control problem statement

To determine a unique point by distance constraints to fixed beacons, we need at least four non-coplanar fixed beacons in three-dimensional space, at least three non-collinear fixed beacons in two-dimensional space, at

least two non-collision fixed beacons in one-dimensional space. More distance constraints than these number is redundant. This motivate us consider the formation whose interaction graph satisfies the following assumption.

Assumption 1: The interaction graph of considered multi-agent system is acyclic persistent and consists of:

- One vertex (called the leader) has no neighbour.
- One vertex (called the first follower) has one neighbour, which is leader.
- One vertex (called the second follower) has two neighbours (leader and first follower); second follower and its neighbours are non-collinear at desired realization.
- One vertex (called the third follower) has three neighbours (leader, first follower and second follower); third follower and its neighbours are non-coplanar at desired realization.
- Every other vertex (ordinary follower) is one of follow-
ing three cases
 - Having two neighbours; ordinary follower and its neighbours are collinear at desired realization.
 - Having three neighbours; ordinary follower and its non-collinear neighbours are coplanar at desired realization.
 - Having four neighbours; ordinary follower and its neighbours are non-coplanar at desired realization.

For a multi-agent system whose interaction topology satisfies Assumption 1, when all inter-agent distance constraints are achieved, the formation of overall group is congruent to the desired one. We assume agents obey the single integrator modelled

$$\dot{\mathbf{p}}_i = \mathbf{u}_i, \forall i \in \mathcal{V} \quad (8)$$

and each agent has information of the set $D_i = \{\bar{d}_{ij} = \|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_j\| : j \in \mathcal{N}_i\} \cup \{\bar{d}_{kl} = \|\bar{\mathbf{p}}_k - \bar{\mathbf{p}}_l\| : (k, l) \in \mathcal{N}_i\}$ where $\bar{\mathbf{p}}$ is the desired realization. We define the desired formation set

$$E_C = \left\{ \mathbf{p} : \|\mathbf{p}_i - \mathbf{p}_j\| = \bar{d}_{ij}, \bar{d}_{ij} \in \bigcup_{i \in \mathcal{V}} D_i \right\} \quad (9)$$

This set is invariant with rotation and translation. We are intent on designing a distributed control law, for which each control law $\mathbf{u}_i$ uses only relative position measurements $\mathbf{p}_j^i, j \in \mathcal{N}_i$, with respect to its own coordinate system, such that agents converge to a realization in E_C .

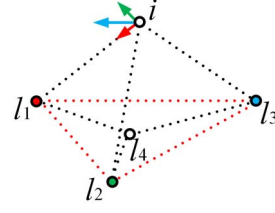


Fig. 3 Ordinary follower with four neighbours in three-dimensional space. The red vector corresponds to $\mathbf{u}_z$, the green one corresponds to $\mathbf{u}_\theta$ and the blue one corresponds to $\mathbf{u}_\phi$.

3.2 Control law

To design control law, we use following functions

$$\mathbf{z}_{ij} = \mathbf{p}_i - \mathbf{p}_j, g_{ij} = 1 - \left(\frac{\bar{d}_{ij}}{\|\mathbf{z}_{ij}\|} \right)^\alpha, 0 < \alpha < 1,$$

$$\mathbf{u}_z(i, l_1) = \begin{cases} \text{sig}(g_{il_1})^\alpha \mathbf{z}_{il_1}, & \text{if } \mathbf{z}_{il_1} \neq \mathbf{0}, \\ \mathbf{v}, & \text{if } \mathbf{z}_{il_1} = \mathbf{0} \end{cases} \quad (10)$$

$$\mathbf{u}_\theta(i, l_1, l_2) = \begin{cases} \left[\frac{[\mathbf{z}_{l_2 l_1} \times \mathbf{z}_{il_1}]}{\|\mathbf{z}_{l_2 l_1} \times \mathbf{z}_{il_1}\|} \times \mathbf{z}_{il_1} \right], & \text{if } [\mathbf{z}_{l_2 l_1} \times \mathbf{z}_{il_1}] \neq \mathbf{0}, \\ \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{z}_{il_1}, & \text{otherwise.} \end{cases} \quad (11)$$

$$\mathbf{u}_\phi(i, l_1, l_2) = \begin{cases} \frac{[\mathbf{z}_{il_1} \times \mathbf{z}_{il_2}]}{2\|\mathbf{z}_{il_1} \times \mathbf{z}_{il_2}\|}, & \text{if } \mathbf{z}_{il_1} \times \mathbf{z}_{il_2} \neq \mathbf{0}, \\ \mathbf{0}, & \text{otherwise.} \end{cases} \quad (12)$$

Vector $\mathbf{u}_z(i, l_1)$ has the same direction as vector $\mathbf{z}_{il_1}$ if i, l_1 are not collided, and is any unit vector $\mathbf{v}$, otherwise. Vector $\mathbf{u}_\theta(i, l_1, l_2)$ is perpendicular to vector $\mathbf{z}_{il_1}$ in the plane $\Delta_{il_1 l_2}$ when i, l_1, l_2 are not collinear, $\mathbf{u}_\theta(i, l_1, l_2)$ is any perpendicular vector to $\mathbf{z}_{il_1}$, otherwise. Vector $\mathbf{u}_\phi(i, l_1, l_2)$ is perpendicular to the plane $\Delta_{il_1 l_2}$ when l_1, l_2 are not collided. We propose control law for each agent

- The leader (agent 1) is stationary

$$\mathbf{u}_1 = \mathbf{0}, \quad (13)$$

- First follower (agent 2) has main task to preserve the distance from it to leader, then the corresponding control law is given as follows

$$\mathbf{u}_2 = -\mathbf{u}_z(2, 1). \quad (14)$$

- Follower i with two neighbours, $\mathcal{N}_i = \{l_1, l_2\}$, needs to maintain two inter-agent distance constraints. Equivalently, it has main task to preserve the distance from it to l_1 and the angle $\angle il_1 l_2$

$$\mathbf{u}_i = -\mathbf{u}_z(i, l_1) - \text{sig}(e_{\theta_i})^\alpha \mathbf{u}_\theta(i, l_1, l_2), \quad (15)$$

where $\bar{\theta}_i = d_\theta(\bar{d}_{il_1}, \bar{d}_{il_2}, \bar{d}_{l_1 l_2})$; $e_{\theta_i} = f_\theta(\mathbf{z}_{il_1}, \mathbf{z}_{il_2}) - \bar{\theta}_i$. We have $0^\circ < \theta_i < 180^\circ$ for the second follower, and $\bar{\theta}_i = 0^\circ$ or $\bar{\theta}_i = 180^\circ$ in the case of ordinary follower.

- Follower i with three neighbours, $\mathcal{N}_i = \{l_1, l_2, l_3\}$, has main task to preserve the distance from it to l_1 , the angle $\angle il_1 l_2$ and the angle $\angle(\Delta_{il_1 l_2}, \Delta_{l_3 l_1 l_2})$.

$$\mathbf{u}_i = -\mathbf{u}_z(i, l_1) - \text{sig}(e_{\theta_i})^\alpha \mathbf{u}_\theta(i, l_1, l_2) - \text{sig}(e_{\phi_i})^\alpha \mathbf{u}_\phi(i, l_1, l_2), \quad (16)$$

where $e_{\theta_i} = f_\theta(\mathbf{z}_{il_1}, \mathbf{z}_{l_2 l_1}) - \bar{\theta}_i$, $\bar{\theta}_i = d_\theta(\bar{d}_{il_1}, \bar{d}_{l_2 l_1}, \bar{d}_{il_2})$, ($0^\circ < \bar{\theta}_i < 180^\circ$), $e_{\phi_i} = f_\phi(\mathbf{z}_{il_1}, \mathbf{z}_{il_2}, \mathbf{z}_{l_3 l_1}, \mathbf{z}_{l_3 l_2}) - \bar{\phi}_i$, $\bar{\phi}_i = d_\phi(\bar{d}_{l_2 l_1}, \bar{d}_{il_1}, \bar{d}_{il_2}, \bar{d}_{l_3 l_1}, \bar{d}_{l_3 l_2}, \bar{d}_{il_3})$. We have $0^\circ < \bar{\phi}_i < 180^\circ$ for the third follower, and $\bar{\phi}_i = 0^\circ$ or $\bar{\phi}_i = 180^\circ$ in the case of ordinary follower.

- Consider follower i with four neighbours, $\mathcal{N}_i = \{l_1, l_2, l_3, l_4\}$. Let $\bar{\phi}_{1i} = d_\phi(\bar{d}_{l_2 l_1}, \bar{d}_{il_1}, \bar{d}_{il_2}, \bar{d}_{l_3 l_1}, \bar{d}_{l_3 l_2}, \bar{d}_{il_3})$, $\bar{\phi}_{2i} = d_\phi(\bar{d}_{l_2 l_1}, \bar{d}_{il_1}, \bar{d}_{il_2}, \bar{d}_{l_4 l_1}, \bar{d}_{l_4 l_2}, \bar{d}_{il_4})$ and $\bar{\theta}_i = d_\theta(\bar{d}_{il_1}, \bar{d}_{l_2 l_1}, \bar{d}_{il_2})$. We have $0^\circ < \bar{\theta}_i, \bar{\phi}_{1i}, \bar{\phi}_{2i} < 180^\circ$.

We select the positive direction of angle $\angle(\Delta_{il_1 l_2}, \Delta_{l_3 l_1 l_2})$ around $l_2 l_1$ from l_4 to l_3 and define the angle $\bar{\phi}_i$ such that $\bar{\phi}_i = \bar{\phi}_{1i}$ if $\bar{\phi}_{2i} - \bar{\phi}_{1i} = \bar{\phi}_{3i}$ or $\bar{\phi}_{1i} + \bar{\phi}_{2i} + \bar{\phi}_{3i} = 360^\circ$, and $\bar{\phi}_i = -\bar{\phi}_{1i}$ otherwise, where $\bar{\phi}_{3i} = d_\phi(\bar{d}_{l_2 l_1}, \bar{d}_{l_3 l_1}, \bar{d}_{l_3 l_2}, \bar{d}_{l_4 l_1}, \bar{d}_{l_4 l_2}, \bar{d}_{il_4})$. The control law of agent i is described as follows

$$\mathbf{u}_i = -\mathbf{u}_z(i, l_1) - \text{sig}(e_{\theta_i})^\alpha \mathbf{u}_\theta(i, l_1, l_2) - \text{sig}(e_{\phi_i})^\alpha h_i \mathbf{u}_\phi(i, l_1, l_2). \quad (17)$$

where $h_i = \text{sign}([[\mathbf{z}_{l_4 l_1} \times \mathbf{z}_{l_4 l_2}] \times [\mathbf{z}_{l_3 l_1} \times \mathbf{z}_{l_3 l_2}]] \cdot \mathbf{z}_{l_2 l_1})$, $e_{\phi_i} = k_{\phi_i} f_\phi(\mathbf{z}_{il_1}, \mathbf{z}_{il_2}, \mathbf{z}_{l_3 l_1}, \mathbf{z}_{l_3 l_2}) - \bar{\phi}_i$, $e_{\theta_i} = f_\theta(\mathbf{z}_{il_1}, \mathbf{z}_{l_2 l_1}) - \bar{\theta}_i$, and k_{ϕ_i} can be calculated by following algorithm

Algorithm 1 Determine k_{ϕ_i} from relative position of neighbours with respect to local coordinate system.

Use equation (7) to calculate

$$\phi_{1i} = f_\phi(\mathbf{z}_{il_1}, \mathbf{z}_{il_2}, \mathbf{z}_{l_3 l_1}, \mathbf{z}_{l_3 l_2}),$$

$$\phi_{2i} = f_\phi(\mathbf{z}_{il_1}, \mathbf{z}_{il_2}, \mathbf{z}_{l_4 l_1}, \mathbf{z}_{l_4 l_2}),$$

$$\phi_{3i} = f_\phi(\mathbf{z}_{l_3 l_1}, \mathbf{z}_{l_3 l_2}, \mathbf{z}_{l_4 l_1}, \mathbf{z}_{l_4 l_2}).$$

if $\phi_{2i} - \phi_{1i} = \phi_{3i}$ or $\phi_{1i} + \phi_{2i} + \phi_{3i} = 360^\circ$ then

$$k_{\phi_i} = 1$$

else

$$k_{\phi_i} = -1$$

end if

When agent i converges to an equilibrium $\mathbf{u}_i = \mathbf{0}$, we have $\|\mathbf{z}_{il_1}\| = \bar{d}_{il_1}$, $\angle il_1 l_2 = \bar{\theta}_i$, $\angle(\Delta_{il_1 l_2}, \Delta_{l_3 l_1 l_2}) = \bar{\phi}_{1i}$, and $\angle(\Delta_{il_1 l_2}, \Delta_{l_4 l_1 l_2}) = \bar{\phi}_{2i}$; or all its inter-agent distance constraints are satisfied.

3.3 Stability analysis

For convenience, we study the dynamic of agent i at finite time τ_i when its neighbours converged to their desired equilibria. If agent i converge to its desired equilibrium in finite time, then the desired formation of overall group is finite-time stable. When agents i, l_1 are collision, control law component $\mathbf{u}_z(i, l_1)$ help agent i escape this point. With $\mathbf{z}_{il_1} \neq \mathbf{0}$, we have

$$\dot{g}_{il_1} = -\left(\frac{\bar{d}_{il_1}}{\|\mathbf{z}_{il_1}\|}\right)^4 \text{sig}(g_{il_1})^\alpha, \dot{e}_{\theta_i} = -\text{sig}(e_{\theta_i})^\alpha, \dot{e}_{\phi_i} =$$

$-\text{sig}(e_{\phi_i})^\alpha$. Consider function $\Phi_i = g_{il_1}^2$, we have

$$\dot{\Phi}_i = -2g_{il_1} \text{sig}(g_{il_1})^\alpha \left(\frac{\bar{d}_{il_1}}{\|\mathbf{z}_{il_1}\|}\right)^4 \leq 0$$

Assume that $\mathbf{p}_i$ converges to the equilibrium $\mathbf{p}_i^*$ where $\|\mathbf{z}_{il_1}\| = 0$. Then there exists a time T such that $\|\mathbf{z}_{il_1}(T)\| > 0$ and agent i converge to $\mathbf{p}_i^*$ through $\mathbf{p}_i(T)$. We have $-\infty < g_{il_1}(\mathbf{p}_i(T)) < 0$ or $\Phi_i(\mathbf{p}_i(T)) < \Phi_i(\mathbf{p}_i^*)$, this contradicts with $\dot{\Phi}_i \leq 0$. So agent i cannot converge to $\mathbf{p}_i^*$.

Let $V_i = |g_{il_1}|^{1+\alpha} + |e_{\theta_i}|^{1+\alpha} + |e_{\phi_i}|^{1+\alpha}$ be the potential function, we have $V_i \geq 0$ and equal to zero if and only if $g_{il_1} = e_{\theta_i} = e_{\phi_i} = 0$, i.e., agent i is at its desired equilibrium. The derivative of V_i with respect to time is given as

$$\begin{aligned} \dot{V}_i &= -(1+\alpha)[\text{sig}(g_{il_1})^{2\alpha} \left(\frac{\bar{d}_{il_1}}{\|\mathbf{z}_{il_1}\|}\right)^4 + \text{sig}(e_{\theta_i})^{2\alpha} + \\ &\quad + \text{sig}(e_{\phi_i})^{2\alpha}] \\ &= -(1+\alpha)\left[\left(\frac{\bar{d}_{il_1}}{\|\mathbf{z}_{il_1}\|}\right)^4 2|g_{il_1}|^{2\alpha} + |e_{\theta_i}|^{2\alpha} + |e_{\phi_i}|^{2\alpha}\right] \\ &\leq -(1+\alpha)[|g_{il_1}|^{2\alpha} + |e_{\theta_i}|^{2\alpha} + |e_{\phi_i}|^{2\alpha}]c_i \end{aligned}$$

where $c_i = \min\left\{\left(\frac{\bar{d}_{il_1}}{\|\mathbf{z}_{il_1}\|}\right)^4, 1\right\}$. The fact $0 < \alpha < 1$ implies $0 < \frac{2\alpha}{\alpha+1} < 1$, from Lemma 2, we have $(|g_{il_1}|^{\alpha+1} + |e_{\theta_i}|^{\alpha+1} + |e_{\phi_i}|^{\alpha+1})^{\frac{2\alpha}{\alpha+1}} \leq |g_{il_1}|^{2\alpha} + |e_{\theta_i}|^{2\alpha} + |e_{\phi_i}|^{2\alpha}$. Then, we have $\dot{V}_i \leq -(1+\alpha)c_i V_i^{\frac{2\alpha}{1+\alpha}}$, according to Lemma 1, V_i converges to 0 in finite time T_i where

$$\begin{aligned} T_i &\leq \frac{(|g_{il_1}(\tau_i)|^{1+\alpha} + |e_{\theta_i}(\tau_i)|^{1+\alpha} + |e_{\phi_i}(\tau_i)|^{1+\alpha})^{1-\frac{2\alpha}{\alpha+1}}}{(1+\alpha)(1-\frac{2\alpha}{\alpha+1})c_i} \\ &\leq \frac{|g_{il_1}(\tau_i)|^{1-\alpha} + |e_{\theta_i}(\tau_i)|^{1-\alpha} + |e_{\phi_i}(\tau_i)|^{1-\alpha}}{(1-\alpha)c_i} \end{aligned}$$

Theorem 1: Consider a multi-agent system of N single-integrator modelled agents, whose interaction network satisfies Assumption 1, under proposed control law given in (13-17), agents converge globally to their desired equilibria, at which the formation of overall group is congruent to the desired formation, in finite time $T = \sum_{i \in \mathcal{V}} T_i$.

3.4 Formation control of agents moving in the plane

In this subsection, we consider the multi-agent system where agents are required to move in the plane. The interaction network of considered group consists of one leader, one first follower, one second follower and ordinary followers (which have two or three neighbours). Control law (15) guarantees agents with two neighbours to converge to the plane containing leader, first follower and second follower. For the follower i with three neighbours, $\mathcal{N}_i = \{l_1, l_2, l_3\}$, it needs to preserve the distance

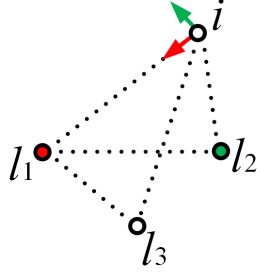


Fig. 4 Ordinary follower with three coplanar neighbours.

from it to l_1 , the angles $\angle il_1l_2, \angle il_1l_3$ as desired values depending on its inter-agent distance constraints and to force the angle $\angle(\Delta_{il_1l_2}, \Delta_{il_1l_3})$ converge to zero. We select the positive direction of angle $\angle il_1l_2$ around l_1 from l_3 to l_2 and define the angle θ_i such that $\bar{\theta}_i = \bar{\theta}_{1i}$ if $\bar{\theta}_{2i} - \bar{\theta}_{1i} = \bar{\theta}_{3i}$ or $\bar{\theta}_{1i} + \bar{\theta}_{2i} + \bar{\theta}_{3i} = 360^\circ$, and $\bar{\theta}_i = -\bar{\theta}_{1i}$ otherwise, where $\bar{\theta}_{1i} = d_\theta(\bar{d}_{il_1}, \bar{d}_{l_2l_1}, \bar{d}_{il_2})$, $\bar{\theta}_{2i} = d_\theta(\bar{d}_{il_1}, \bar{d}_{l_3l_1}, \bar{d}_{il_3})$, $\bar{\theta}_{3i} = d_\theta(\bar{d}_{l_2l_1}, \bar{d}_{l_3l_1}, \bar{d}_{l_3l_2})$. The control law of agent i is described as follows

$$\mathbf{u}_i = -\mathbf{u}_z(i, l_1) - \text{sig}(e_{\theta_i})^\alpha h_i \mathbf{u}_\theta(i, l_1, l_2) - \text{sig}(e_{\phi_i})^\alpha \mathbf{u}_\phi(i, l_1, l_2) \quad (18)$$

where $h_i = \text{sign}([\mathbf{z}_{l_3l_1} \times \mathbf{z}_{l_2l_1}] \cdot [\mathbf{z}_{l_2l_1} \times \mathbf{z}_{il_1}])$, $e_{\phi_i} = f_\phi(\mathbf{z}_{il_1}, \mathbf{z}_{il_2}, \mathbf{z}_{l_3l_1}, \mathbf{z}_{l_3l_2})$, $e_{\theta_i} = k_{\theta_i} f_\theta(\mathbf{z}_{il_1}, \mathbf{z}_{l_2l_1}) - \bar{\theta}_i$ and k_{θ_i} can be calculated by following algorithm

Algorithm 2 Determine k_{θ_i} from relative position of neighbours with respect to local coordinate system.

Use equation (6) to calculate

$$\theta_{1i} = f_\theta(\mathbf{z}_{il_1}, \mathbf{z}_{l_3l_1}, \mathbf{z}_{il_3}),$$

$$\theta_{2i} = f_\theta(\mathbf{z}_{il_1}, \mathbf{z}_{l_2l_1}, \mathbf{z}_{il_2}),$$

$$\theta_{3i} = f_\theta(\mathbf{z}_{l_2l_1}, \mathbf{z}_{l_3l_1}, \mathbf{z}_{l_2l_3}).$$

if $\theta_{2i} - \theta_{1i} = \theta_{3i}$ or $\theta_{1i} + \theta_{2i} + \theta_{3i} = 360^\circ$ **then**

$$k_{\theta_i} = 1$$

else

$$k_{\theta_i} = -1$$

end if

Theorem 2: Consider a multi-agent system of N single-integrator modelled agents, whose interaction network consists of leader, first follower, second follower, ordinary with two or three neighbours in Assumption 1, under proposed control law given in (13, 14, 15, 18), the desired formation of overall group is globally finite-time stable and agents will not escape the plane containing leader, first and follower after they converge to this plane.

4. SIMULATION

4.1 Formation moving freely in three-dimensional space

We provide the simulation to demonstrate our analysis. Let the desired distances are $\bar{d}_{21} = \bar{d}_{31} =$

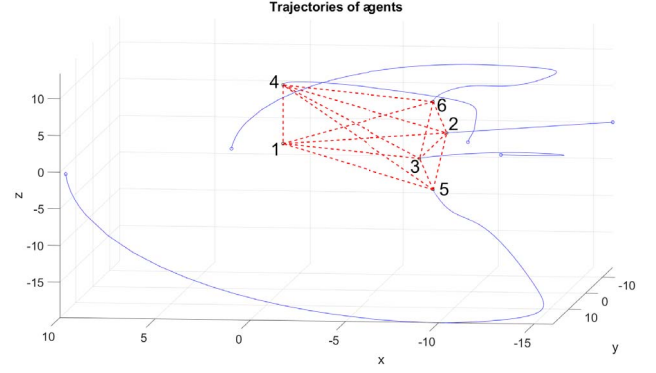


Fig. 5 The desired formation is achieved in a finite time.

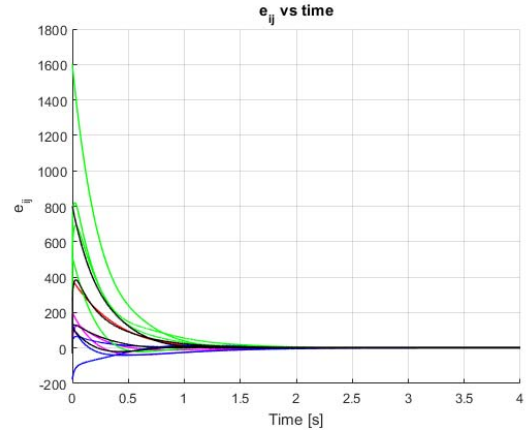


Fig. 6 The square distance errors converge to zero.

$8\sqrt{2}, \bar{d}_{32} = 16, \bar{d}_{41} = 8, \bar{d}_{42} = \bar{d}_{43} = 8\sqrt{3}, \bar{d}_{51} = \bar{d}_{52} = \bar{d}_{53} = 10, \bar{d}_{54} = 2\sqrt{65}, \bar{d}_{61} = \bar{d}_{62} = \bar{d}_{63} = 10, \bar{d}_{64} = 8$. From the initial condition, $\mathbf{P}_0 = [\mathbf{p}_1^T(0), \mathbf{p}_2^T(0), \mathbf{p}_3^T(0), \mathbf{p}_4^T(0), \mathbf{p}_5^T(0), \mathbf{p}_6^T(0)]^T = [0, 0, 0, -16, -16, 0, -12, 5, 0, -10, 2, 1, 10, 16, -1, 2, 8, 1]^T$, agents converge to their desired equilibria in finite time under designed control law (Fig. 5 and Fig. 6).

4.2 Formation moving limited in the plane

We consider the system of five agents which are required to move in the plane. The desired distances are given as follows $\bar{d}_{21} = \bar{d}_{31} = 4\sqrt{2}, \bar{d}_{32} = 8, \bar{d}_{41} = 2\sqrt{29}, \bar{d}_{42} = 10, \bar{d}_{43} = 6, \bar{d}_{52} = 6, \bar{d}_{53} = 10, \bar{d}_{54} = 8$; and the initial condition is $\mathbf{P}_0 = [\mathbf{p}_1^T(0), \mathbf{p}_2^T(0), \mathbf{p}_3^T(0), \mathbf{p}_4^T(0), \mathbf{p}_5^T(0)]^T = [0, 0, 0, 12, 12, -5, -10, 2, 10, 16]^T$. Fig. 7 and Fig. 8 provide the simulation for the finite-time stability of desired formation.

5. CONCLUSION

In this paper, we have studied the formation control of multi-agent systems whose interaction graphs are acyclic generically persistent. The proposed control law uses only relative position measurements with respect to local frames and the coordinate systems of agents are not

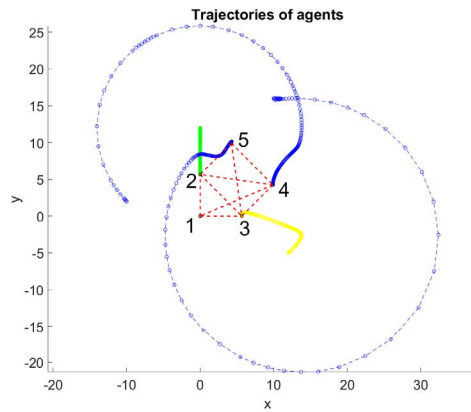


Fig. 7 The desired formation is achieved in a finite time.

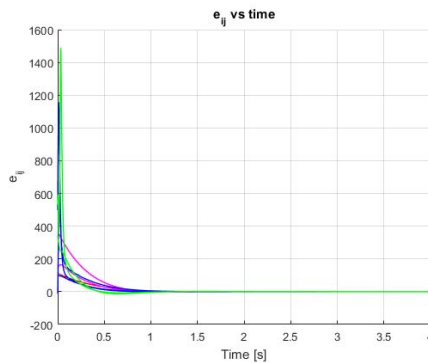


Fig. 8 The square distance errors converge to zero.

required to align together. The trajectories of agents are guarantee successfully to converge globally to their desired equilibria, where the shape of overall group is congruent to the desired formation, in finite time.

ACKNOWLEDGEMENT

This work was supported by the Korea Electric Power Corporation through Korea Electrical Engineering & Science Research Institute under Grant R15XA03-00 and by the National Research Foundation (NRF) of Korea under the grant NRF-2017R1A2B3007034.

REFERENCES

- [1] R. Grabowski, Luis E. N-S, C. J.J. P. K. Khosla, "Heterogeneous Teams of Modular Robots for Mapping and Exploration", *Autonomous Robots* 8, 293308, 2000
- [2] W. Burgard, M. Moors, D. Fox, R. Simmons, S. Thrun, "Collaborative multirobot exploration", *Proc. of the 2000 IEEE conference on robotics and automation*, pp. 476–481.
- [3] N. Shiroma, Y. Chiu, N. Sato, F. Matsuno, "Cooperative task execution of a search and rescue mission by a multi-robot team", *Advanced Robotics*(3), 2005, pp. 311–329.
- [4] K.-K. Oh, M.-C. Park, H.-S. Ahn, "A survey of multi-agent formation control", *Automatica*, Vol.53, 2015, pp. 424–440.
- [5] L. Krick, M. E. Broucke, B. A. Francis, "Stabilisation of infinitesimally rigid formations of multi-robot networks", *International Journal of Control*, Vol. 82, 2009, pp. 423–439.
- [6] D. V. Dimarogonas, K. H. Johansson, "Stability analysis for multi-agent systems using the incidence matrix: quantized communication and formation control", *Automatica*, Vol. 46, 2010, pp. 2379–2384.
- [7] K.-K. Oh, H-S Ahn, "Distance-based undirected formations of single- and double-integrator modelled agents in n-dimensional space", *Int. J. Robust and Nonlinear Control*, Vol. 24, No. 12, 2014, pp. 1809–1820.
- [8] M. Cao, B. D. O. Anderson, A. S. Morse and C. Yu, "Control of acyclic formation of mobile autonomous agents", *Proc. IEEE Conference on Decision and Control*, 2008, pp. 1187–1192.
- [9] M-C. Park, K-Jeong, H-S. Ahn, "Formation stabilization and resizing based on the control of inter-agent distances", *International Journal of Robust and Nonlinear Control*, Vol. 25, No. 14, 2015, pp. 2532–2546.
- [10] M.-C. Park, Z. Sun, K.-K. Oh, B. D. O. Anderson, H.-S. Ahn, "Finite-time convergence control for acyclic persistent formations", *Proc. in IEEE International Symposium on Intelligent Control (ISIC) Part of 2014 IEEE Multi-conference on Systems and Control October 8-10, 2014*.
- [11] Z. Sun, S. Mou, M. Deghat, B. D. O. Anderson, "Finite time distributed distance-constrained shape stabilization and flocking control for d-dimensional undirected rigid formations", *Int. J. Robust Nonlinear Control*, 2015.
- [12] G. Hardy, J. Littlewood, G. Polya, "Inequalities", *Cambridge: Cambridge University Press*, 1952.
- [13] Bhat SP, Bernstein DS, "Finite-time stability of continuous autonomous systems", *SIAM Journal on Control and Optimization* 2000; 38(3), 751–766.
- [14] C. Yu, J. M. Hendrickx, B. Fidan, B. D. O. Anderson, V. D. Blondel, "Three and higher dimensional autonomous formations: Rigidity, persistence and structural persistence", *Automatica*, Vol. 43, 2007, pp. 387–402.
- [15] J. M. Hendrickx, B. D. O. Anderson, J. Delvenne, V. D. Blondel, "Directed graphs for the analysis of rigidity and persistence in autonomous agent systems", *Int. J. Robust Nonlinear Control*, vol. 17, no. 10-11, 2007, pp. 960–981.
- [16] V. H. Pham, M. H. Trinh, H-S. Ahn, "Finite-time convergence of acyclic persistent formations in the plane", *Submitted to IEEE Conference on Decision and Control*, 2017.